# Morphophonology

## *Topic 4: Less (and more) morphological structure than expected*

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# Raffelsiefen (2007) on *sub-*

- Claimed contrast (slightly augmented by AA)

| | Casual | Careful |
|---|---|---|
| *subordinate* | səˈbɔɹdənət | sʌˈbɔɹdənət |
| *suborbital* | sʌbˈɔɹbəɾl̩ | sʌbˈʔɔɹbəɾl̩ |

- Raffelsiefen: [subordinate] vs. sub[orbital]
- Or cyclic evaluation: /sʌb+[ʔɔɹbəɾl̩]/ (?)
- OO-Faithfulness: base [ʔɔɹbəɾl̩]

# *suborbital* vs. *subordinate*

- Why this difference?
- Hypothesis: phonological difference mirrors a morphological difference
  - Two different *sub-* prefixes, with different prosodification or derivation
  - Prefixed vs. pseudoprefixed
  - "Degrees" of complexity?
- Learned distinction? By-product of something?

# Not just *sub*-! English aspiration

- Reminder of the basic contrast

| Tautomorphemic sT | Prefixed s-T |
|---|---|
| dis[p]lay | dis-[p$^h$]lease |
| des[p]otism | dis-[p$^h$]ossess |

- Aspiration not blocked in sC across prefix/stem boundary

# Variable aspiration after prefixes

- A surprising contrast among prefixed forms

| Aspirated | Unaspirated |
|---|---|
| ex-[$k^h$]ommunicate | ex-[p]lode |
| dis-[$p^h$]lease | mis-[t]ake |
| dis-[$p^h$]ossession | pre-[p]osition |
| dis-[$k^h$]ontent | dis-[k]ourage |

- Zuraw and Peperkamp (2015)
- Hypotheses about relevant factors?

# Zuraw and Peperkamp (2015)

Selected examples:

| Word | C | ? | $C^h$ | % $C^h$ |
|---|---|---|---|---|
| mistrial | 0 | 0 | 14 | 100% |
| misquote | 0 | 0 | 16 | 100% |
| discontent | 2 | 0 | 14 | 88% |
| discredit | 4 | 0 | 12 | 75% |
| discourtesy | 5 | 3 | 7 | 58% |
| displaced | 8 | 3 | 5 | 38% |
| distant | 13 | 0 | 3 | 19% |
| mistaken | 11 | 3 | 2 | 15% |
| discretion | 14 | 1 | 1 | 7% |
| distract | 13 | 3 | 0 | 0% |
| disturb | 14 | 2 | 0 | 0% |
| dispel | 16 | 0 | 0 | 0% |

# Northern Italian s-voicing (Baroni 2001)

- Recall pattern: intervocalic s-voicing blocked at prefix+stem boundaries

| | | | |
|---|---|---|---|
| a. | /isola/ | izola | 'island' |
| | /kaʊs-a-v-a/ | kaʊzava | 'he/she was causing' |
| | /famos-isːim-o/ | famozisːimo | 'very famous' |
| b. | /ri-sal-a-re/ | risalare | 're-salt' |
| | /porta-sigaretːa/ | portasigaretːa | 'cigarette case' |
| | /lavando-si/ | lavandosi | 'washing oneself' |
| | /la sera/ | lasera | 'the evening' |
| | /bɛlːa sera/ | bɛlːasera | 'beautiful evening' |

# …but not all prefix+stem boundaries?

- Like English mis-[t]ake, some stem-initial /s/ in Italian show phonotactically expected voicing

| | | | |
|---|---|---|---|
| pre-seleˈtːsjone | 'preselection' | de-ˈzerto | 'desert' |
| pre-seˈnile | 'presenile' | pre-ˈzadʒo | 'premonition' |
| a-soˈtʃale | 'asocial' | pre-ˈzunto | 'presumed' |

# Baroni (2001)

- 102 existing words, with various prefixes
  - anti-, a-, bi-, de-, para-, poli- pre-, ri-, etc.
- Items varied in terms of
  - Semantic transparency
    - risuscitare 'resurrect': ri+suscitare 'arouse'
      risalto 'prominence': ri+salto 'jump'
  - Frequency of base form
  - Length of prefix

# Baroni (2001): Results

- Some items 100% [z], some items 100% [s]
- Many items fall somewhere in between

| Word | Gloss | % [s] | N [s] | Total |
|---|---|---|---|---|
| risalto | prominence | 100% | 16 | 16 |
| bisettrice | bisecting | 93% | 26 | 28 |
| riserva | reserve | 91% | 50 | 55 |
| risorto | resurrected | 91% | 48 | 53 |
| trasognata | dreamy | 89% | 48 | 54 |
| preside | principal | 84% | 49 | 58 |
| presalario | stipend | 83% | 50 | 60 |
| resiliente | resilient | 76% | 16 | 21 |
| bisesto | leap-year | 68% | 21 | 31 |
| presuntivo | estimate | 0% | 0 | 0 |
| resistente | resistant | 0% | 0 | 0 |
| risultato | result | 0% | 0 | 0 |

# Statistical analysis

- Many possible predictors: frequency and semantic transparency of base, transparency of prefix, etc.
- Many are correlated and hard to disentangle, but two factors emerge clearly
- Length of the prefix
    - Voicing blocked with longer prefixes (*anti-, para-, poli-*)
    - E.g., *polisindeto*, *antisettico*
- Semantic transparency of the root

# Baroni's claim

- Length of prefix and semantic transparency encourage learners to segment words and store them as morphologically complex
- Phonology can only treat the item as morphologically complex if the speaker has stored it as such
- Baroni (2000): MDL model of morphological segmentation

# The lesson that's emerging

- Cases like English *mistake* and *disposition* aren't just sporadic exceptions
- Arise when etymologically complex words are learned as units
- If our theory of how morphology influences phonology invokes morpheme boundaries, we need a serious theory of morphological segmentation
- Many theories on the market (MDL, Bayes) "undersegment", from an etymological point of view

# Tagalog tapping (Zuraw 2006)

- Allophonic alternation between d ~ ɾ
- Allophonic distribution

| | | | | | |
|---|---|---|---|---|---|
| dalaˈwa | 'two' | ˈʔaɾaw | 'sun' | laˈkad | 'walk' |
| daˈtiŋ | 'arrive' | ˈpuɾi | 'praise' | liŋˈkod | 'servant' |
| diˈlaw | 'yellow' | haˈɾap | 'front' | liˈkod | 'back' |

- Alternations

| | | | | |
|---|---|---|---|---|
| laka[d] | 'walk' | ~ | laka[ɾ]an | 'be walked on' |
| [d]umi | 'dirt' | ~ | ma[ɾ]umi | 'dirty' |

# Variation in tapping

- Not all prefixed words tap as expected
- Variability within and across speakers

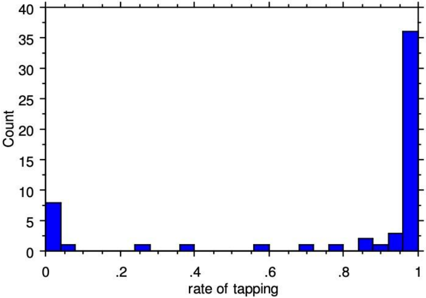
| | | | | |
|---|---|---|---|---|
| [d]umi | 'dirt' | ~ | ma[ɾ]umi | 'dirty' |
| [d]ahon | 'leaf' | ~ | ma[d]ahon | 'leafy' (*ma[ɾ]ahon) |

# Zuraw's corpus study

- Searched for prefixed words, in corpus of ~20 million Tagalog words scraped from web
- Prefixed words: attested Tagalog words from the dictionary, with /d/-initial stems
- Searched both tapped and untapped variants
- Analysis focuses first on words with total frequency ≥ 10 (ample opportunity for variation)
  - Examine trends for rare words separately

# Zuraw (2006) results

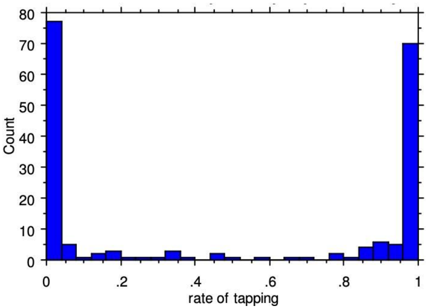
- Tapping robustly attested, but many hits for untapped variants
- Prefixed words that are more frequent than their corresponding bases: strongly prefer tapping




(Zuraw 2006, Fig. 3)

# Zuraw (2006) results

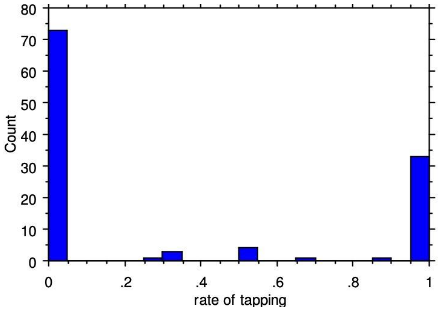
- Prefixed words that are less frequent than their corresponding bases: weakly prefer not to undergo tapping




(Zuraw 2006, Fig. 4)

# Zuraw (2006) results

- Among rare words (freq ≤ 10), tapping even more strongly dispreferred
  - These are generally less frequent than corresponding base




(Zuraw 2006, Fig. 5)

## Zuraw's interpretation: race-based (de-)composition

- Assume speakers have two options when learning/parsing/producing complex expressions
  - Compose from parts:  /ma/ + /dahon/
  - Access/store as a whole: /madahon/
- Speed of access depends on frequency
- Speed of (de-)composition depends on
  - Frequency of parts
  - Additional time to (de-)compose
- Low freq affixed words: often faster to (de-)compose
- High freq affixed words (esp. relative to base): whole form access wins the race

# Decomposed vs. morphologically complex

- "Accessed as a whole" ≠ "fail to parse"
- Consistent with storing "pre-compiled" complex expressions
- Accessed as a whole = no cyclic evaluation, no transderivational base, no derivational stem(?)

Zuraw: "I will follow Baroni loosely in assuming that words like marumi are accessed as a single lexical unit (without taking a position on whether that lexical entry contains information about morpheme boundaries)."

# Shape of an analysis

- Assume [±tap] feature distinguishes [d] from [ɾ]

| | [±tap] |
|---|---|
| d | + |
| ɾ | - |

- Flapping: *VdV >> IO-Ident([±flap])

| /dumi/ | *VdV | IO-Ident ([±tap]) |
|---|---|---|
| a. ☞ dumi | | |
| b. ɾumi | | *! |

| /ma-dumi/ | *VdV | IO-Ident ([±tap]) |
|---|---|---|
| a. madumi | *! | |
| b. ☞ maɾumi | | * |

# Underdecomposition

"Blocking constraints"

- Align-L
- OO-Ident([±tap])

No base/stem *dumi*

| /ma-dumi/ | "Blocker" | *VdV | IO-Ident([±tap]) |
|---|---|---|---|
| a. madumi | | *! | |
| b. ☞ marumi | | | * |

| /ma-dahon/<br>Contains /dahon/ | "Blocker" | *VdV | IO-Ident([±tap]) |
|---|---|---|---|
| a. ☞ madahon | | * | |
| b. marahon | *! | | * |

(What is the blocking constraint in a cyclic account?)

# A pervasive phenomenon

- These cases are not unusual
  - English aspiration: *mistake*, *preparation*, *preposition*
  - English regressive nasalization: *highness* vs. *shyness*
  - Catalan *sub-* (*su*[p]*aquàtic*, *su*[p]*especific* vs. *su*[β]*ordinat*, *su*[p]*scriure*)
- Though not universal
  - German: frequent and opaque *er*[ʔ]*innern* 'remember', etc.

# Compositional vs. whole-form access

- O’Donnell (2015): Bayesian model of morphological composition (following approach of Goldwater et al. 2009)
- Baroni (2000): Minimum Description Length
- These approaches ground compositional vs. whole form behavior in a model of learning, processing
- Cross-linguistic variability is surprising

# Composition vs. complexity

- O'Donnell, Zuraw: forms may be accessed as a whole, but this does not preclude morphologically complex representations
  - cf. Marantz (1995), etc.
- Whole-form access
  - Missing cycles?
  - No access to base ⇒ no faithfulness?

# Surprising complexity

- German *er-innern* 'remember'
- English stress on "pseudo-prefixed" verbs
  - *dismay*, *distend*, etc.
- Pseudo-affixation
  - English stress in *honest* (Hammond 1999)
  - Pseudoreduplication (Zuraw 2002)
  - Pseudosuffixation in processing (Longtin et al. 2003, Rastle et al. 2004; but cf. Rueckl and Aicher 2008)

# Surprising complexity

- Factors beyond frequency and transparency
    - Morphological productivity
    - Phonotactics: stress, clusters, etc.
    - Length
- A current gap: relation between phonological and psycholinguistic evidence for decomposition

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